

DEDA outcomes

We developed an ML-based dropout-risk advice tool for a higher education institution. The goal is to identify students who may be at risk of dropping out early, so student counsellors can provide timely, personalised support (tutoring, mental health guidance, financial support). The model output is advisory only and never makes automatic decisions.

Data and privacy

We use an anonymised dataset, but we recognise that anonymised data can still contain sensitive proxies (e.g., socio-economic and parental background). Therefore, we treat these variables carefully, minimise unnecessary data use, and focus on preventing misuse of the output. Access to the model results is strictly limited to authorised student counsellors, teachers do not have access to reduce stigma and the risk of biased treatment.

Fairness and bias mitigation

Because protected characteristics or proxies (e.g., nationality/international status, gender, age, special needs indicators, socio-economic variables) can lead to unequal outcomes, we assessed representation across student groups. We applied instance weighing so that different groups have equal weight during training.

Transparency, interpretability, and accountability

Predictions are treated as signals, not conclusions. The output should be understandable for counsellors, with a clear explanation of the main contributing factors. Human judgement remains central: counsellors decide whether and how to follow up, using documented guidelines and a consistent process. Accountability is organised by assigning clear responsibilities: developers maintain the model, counsellors interpret and act on advice, and the institution monitors ethical and lawful use.

Monitoring and governance

We emphasise continuous monitoring after deployment, including periodic evaluation of accuracy, fairness, and unintended effects. If the model performs poorly, produces unfair outcomes, or causes harm, an exit strategy should allow the institution to stop or redesign the system.

Audit conclusion

Through the FRAIA certain model decisions are made but not in other documents

Security: stop button. Exit strategy, how it works(expansion and when). Mention when the model will be themed and how.

Algorithm is context specific (add in the FRAIA who we are in and what we do)

Impact assessment: data is vulnerable, tangible effects that could have but not on what those risks are or what it means when someone is going to the student counsellor. What will happen to the students when you are flagged as at risk.

Responsibility: who is responsible, how it will reflect back on people who made it (who is responsible for the algorithm)

Strategy on less traditional variables

Dataset is not enough

Questionnaires for who dropout and not and then get the variables (when you would do the algorithm in real life)

How will you measure the effect of the algorithm

What measures would you normally use

How are the insights of the DEDA adapt to the changes of editing datasources
the choices of it

You had DEDA and FRAIA what it does to your algorithm

Audit:

How did you do the audit and look at the documentation

Then

How did you implement it (the feedback)

How did you edit your technical parts from the DEDA and FRAIA

Address the shortcomings

How would accountability be done